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ORIGINAL RESEARCH

A deep reinforcement learning framework for UAV navigation and selection of relay paths

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Abstract The given paper proposes a deep reinforcement learning framework for unmanned aerial vehicles (UAVs) navigation to provide optimal communication for the users. It comprises gated recurrent unit enhanced with graph attention mechanism to make effective use of inter-UAV communication network for enhanced data retrieval and decisions. To further improve exploration and robustness, the proposed framework is trained using maximum-entropy reinforcement learning, enabling UAVs to learn stochastic policies. A heuristic reward function, solely based on local observations, is designed to optimize global performance in terms of coverage, fairness, and energy efficiency. Extensive simulations validate the effectiveness of the approach, showing superior scalability, adaptability, and communication efficiency over existing methods.

Keywords UAVs · GAT · GRU · DRL · Optimization · Relay paths

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1 Introduction

With the advancement of cutting-edge technologies such as on-board computing systems, sophisticated flight control frameworks, and Ad-hoc networks, managing large-scale, intelligent UAV swarms for complex operations is now possible. Currently, UAV swarms are being used in various domains such as surveillance, logistics, infrastructure inspection, and disaster response. A significant application includes UAV-mounted mobile base stations which can be used to extend communication coverage and performance in areas with insufficient infrastructure. Mobile base stations have the advantage of rapid and flexible deployment, especially in emergencies such as natural disasters that cause disruption to cellular communication networks. In recent years, UAV's have been recognized as cost-effective alternatives that can deliver high-quality communication services. The communication is facilitated among the UAVs while it allows access to the ground users. The limitations such as short communication ranges and high mobility do not favour the use of the centralized control approach [1, 2]. Moreover, since obtaining full global data is usually infeasible in real-world applications, training samples collected are only information collected by individual UAVs and their immediate neighbors. This is different from previous approaches, which utilize global data during training to stabilize model convergence [3, 4]. It is typically infeasible to obtain global information in real tasks; we assume that the training samples of the model only consist of the information from itself and its interconnected neighbors. In other words, the policy is trained with only local information yet should achieve high performance in terms of the global metric [5, 6]. The present work addresses partial observability through a flying Ad-hoc network, where links of interconnected UAVs within the given communication range are formed and

used for low-latency communication. This communication strategy of neighbouring UAVs helps alleviate this partial observability at the most minimal cost thereby developing the concept of GRU and GAT.

1.1 Key contributions

- The given paper introduces a DRL framework to provide effective communication among UAV's. It comprises GAT + GRU for spatial data extraction and a memory unit for data retention.
- Self-attention mechanism using GAT enables the learning of stochastic policies through the proposed model. In GAT, each UAV is represented as a node within a graph, with node embedding. These embedding are shared via edges between nodes, enabling each UAV to process both its own and neighbouring nodes' information.
- This approach ensures that all UAV's follow the same policy for decentralized path planning and control, utilizing ad-hoc network communication.
- Extensive simulations validate the effectiveness of heuristic reward function. Additionally, this study examines various aspects of the proposed framework, including its performance, scalability, transferability, robustness, and interpretability.

1.2 Organization

The remainder of this paper is structured as follows: Sect. 2 discusses related works. Section 3 outlines the system model and the proposed approach. Section 4 deals with the results. Finally, Sect. 5 concludes the study followed by references.

2 Related works

Recent research [7] has extensively explored UAV deployment to enhance its practicality in real-world applications. Some studies have proposed multi-UAV control models that optimize deployment coverage while maintaining energy efficiency. Approaches have also been proposed for reducing the deployment delay and total operational latency. Furthermore, architectures have been devised for optimal deployment and mobility of multiple UAVs, thereby making efficient use of energy resources on the ground equipment. With the development of Ad-hoc networking, in which all nodes that fall within a certain distance can easily form contacts, researchers have aimed at multi-UAV control strategies that reasonably assume that UAVs can maintain communication with their neighbors without even slight delay. When operating a UAV swarm in unobserved environments with low observation levels, an efficient exploration strategy is vital. Among the most popular strategies is the greedy strategy [8], which encompasses elements of introducing random action selection to enhance exploration while refining the decision-making process step by step. Several advanced methods have been proposed for multi-UAV cooperative exploration. These include game-theoretic frameworks for collaborative search and monitoring, cooperative strategies specifically designed for UAV hybrid systems to minimize exploration distance that reduces data exchange by sharing only essential frontier points [9]. Various multi-objective optimization algorithms have been introduced to assign UAV tasks and plan routes efficiently, incorporating genetic algorithms to minimize processing time. Such algorithms lack mathematical aspects like coverage, observation radius, velocity vector and favourable points based on number of vehicles. Other approaches rely on station-based information to determine optimal paths for leading UAVs. The station-based information requires graphical validation which is not possible with existing approaches. Further, some researchers have applied mean-field game control theory to achieve precise positioning while reducing flight energy consumption [10]. More recent studies have adopted reinforcement learning-based strategies to optimize UAV swarm navigation, using lightweight policies with minimal computational overhead. DRL-based models seek to control each UAV independently, leveraging advancements in multi-agent deep reinforcement learning [11]. This approach has been used to develop UAV controllers for formation control and small fixed-wing UAV flocking, incorporating parameter-sharing techniques to accelerate learning convergence. The parameters need to be evaluated based on communication range and observation radius. To address instability issues in multi-agent environments, researchers have introduced centralized-decentralized training where decentralized UAV policies are trained using a centralized critic network that has access to global state information [12]. This method has been applied to optimize UAV trajectories for secure communication and to improve the efficiency of information aggregation through self-attention mechanisms. Thus, in order to alleviate above issues, the proposed framework is designed to ensure adequate communication coverage under varying network conditions.

3 System description

Following parameters are used to define the proposed framework:

Maximum communication range of each UAV = M_{comm}
Each UAV operates at fixed height = H
Coverage Radius = R_c
Observation radius = R_o
Time slot = T
Observation space = OS
Velocity vector = V

Favourable points = FP
Number of UAV's = N

3.1 Evaluation metrics

To assess UAV swarm performance, three key evaluation metrics are introduced: coverage, fairness, and energy consumption. The objective is to optimize these metrics collectively.

- **Coverage index (CI)** [13]: this metric measure the proportion of FP's covered by at least one UAV over the past T time slots. A FP is classified as "covered" if it falls within the communication range of any UAV. The coverage index at a given time slot t is calculated in Eq. (1):

$$CI = Swt(k)St \tag{1}$$

where S represents the total number of FPs, and wt (k) denotes the number of time slots during which FP is covered.

- **Energy consumption (E)** [14]: UAV energy expenditure is assumed to be proportional to flight distance. The energy consumption per UAV at time slot T is given in in Eq. (2):

$$E = e0 + k \times D \tag{2}$$

where e0 = 0.5e_0 = 0.5 represents the energy required for hovering, D is the normalized flight distance, and k is a coefficient set to 0.5. The total energy index across N UAVs and T time slots is defined in Eq. (3):

$$E = 1tN \sum \tau \tag{3}$$

To integrate these metrics, a combined **Coverage-Fairness-Energy (CFE) score** [15] is introduced in in Eq. (4):

$$CFE = CI \times E \tag{4}$$

A higher **CFE score** indicates better performance, as it signifies increased coverage, balanced fairness, and efficient energy utilization.

3.2 Proposed mechanism

The proposed DRL framework (GRU + GAT) optimizes the placement of the UAV and the selection of relay paths under varying coverage environment. This approach ensures that all UAV's follow the same policy for decentralized path planning and control, utilizing ad-hoc network communication. The proposed framework comprises encoder for processing input features, GAT and GRU (memory unit) as shown in Fig. 1.

Initially, the framework employs an encoder to process raw input in Eq. (5):

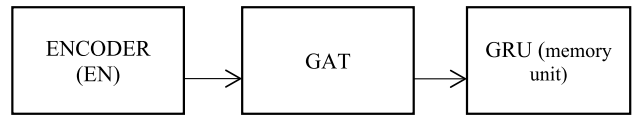

Fig. 1 Blocks of the proposed framework

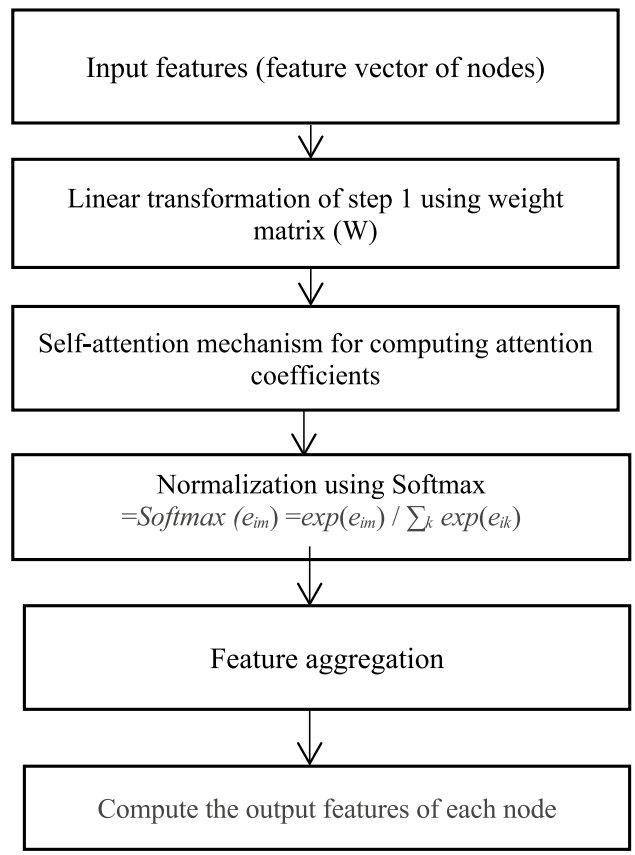

Fig. 2 Working of GAT

$$\mathrm{EN} = \mathrm{R_o(OS)} \tag{5}$$

where OS represents the local observation and R_o is observation radius by UAV. To embed the communication protocol within the network, GAT is incorporated. In GAT, each UAV is represented as a node within a graph, with node embedding. These embedding are shared via edges between nodes, enabling each UAV to process both its own and neighbouring nodes' information. Figure 2 shows the working of GAT.

- Self-attention mechanism for computing attention coefficients is represented using Eq. (6) as:

$$A_{ij} = softmax_{\,j}\left(\frac{Qi_i \cdot K_{jj}^T}{\sqrt{d_k}}\right) \tag{6}$$

where:

$Q_i = W_Q e_i$—query vector for node i
$K_j = W_k e_j$—key vector for node j
d_k—dimensionality of the key vectors
$softmax_j$ — normalization across all neighbors of node i

- Graph Attention Update

$$h_i = \sum_{j \in N(i)} \#_{ij} \cdot V_j \tag{7}$$

where:
$V_j = W_V e_j$– value vector of neighbor j
$N(i)$ — neighborhood of node i
$\#_{ij}$ — attention coefficient weighting neighbor j

- Temporal GRU Integration is represented as:

$$h_{i,t} = GRU\left(h_i, h_{i,t-1}\right) \tag{8}$$

where:
h_i—static embedding from GAT at time t
$h_{i,t-1}$—previous hidden state of node i
$h_{i,t}$—updated temporal embedding.

Figure 3 shows GRU structure used in the proposed framework.

4 Results and discussions

The proposed framework is comprehensively tested to ensure it is scalable, robust, and accurate in navigation. The test was orchestrated to compare with state-of-the-art deep reinforcement learning (DRL) frameworks and conventional UAV navigation techniques across different conditions. All tests were carried out in a multi-UAV setup where 20 UAVs were used to train followed by testing across different scales (5 to 40 UAVs) to assess generalization. Each setting was simulated for 100 independent runs, and the results reported are averaged values.

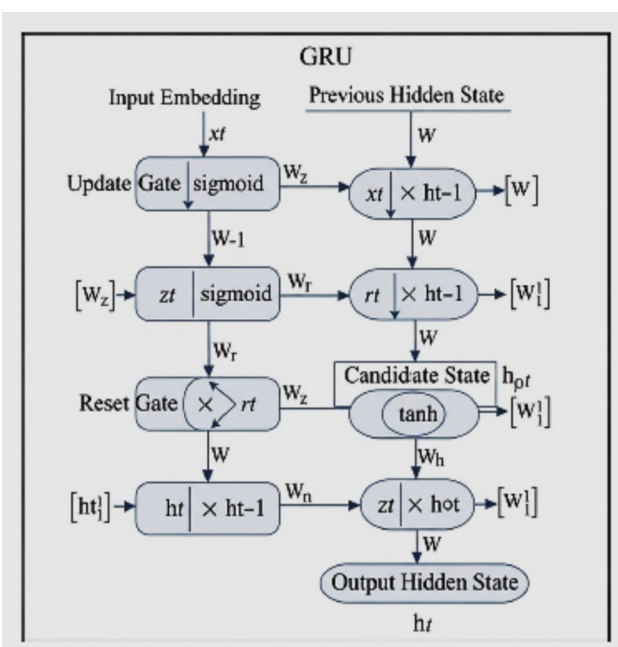

Fig. 3 GRU structure (h_t denotes current hidden state, h_t' denotes previous hidden state, W denotes weight matrix)

- *Scalability analysis*

For measuring scalability, Coverage-Fairness-Energy (CFE) score is chosen as the major metric. Figure 4 presents how CFE values change with the number of UAVs. The results are compared with baseline models such as DRGN (deep recurrent graph network), DGN (deep graph network), MAAC (message authentication acceleration protocol), DQN (deep Q network), and CommNet (communication network). Classic DRL algorithms like DGN and DRGN exhibit reasonable flexibility as the number of UAVs increases, but performance drops precipitously for more than 30 UAVs because of communication bottlenecks. Traditional baselines like DQN, CommNet, and MAAC degrade further, especially at high UAV density, with CFE scores dropping below 70 in most cases.

Conversely, the GRU + GAT approach in this work maintains consistently high CFE scores across scales, up to a level of over 90 in the 40-UAV case. This robustness is indicative of the capability of attention-based semantic communication combined with temporal modeling to process larger fleets with little degradation.

- *Robustness against communication drop rate*

Another important assessment was performed by altering the drop rate of communication, emulating packet loss and unreliable channel conditions that are commonly found in actual UAV deployments. Figure 5 illustrates that increasing drop rate makes all models decrease their CFE scores. Nevertheless, the introduced framework has much slower degradation, with CFE values exceeding 80 even at high drop rates. In comparison, DRGN and DGN fall in the 70–75 categories, whereas baseline methods such as CommNet and DQN drop below 65. Such robustness validates that the graph attention mechanism successfully compensates for lost communication connections by focusing on the most important neighbors.

4.1 Navigation accuracy and error rate comparison with existing studies

In order to compare navigation accuracy, the presented GRU + GAT paradigm was compared to six popularly

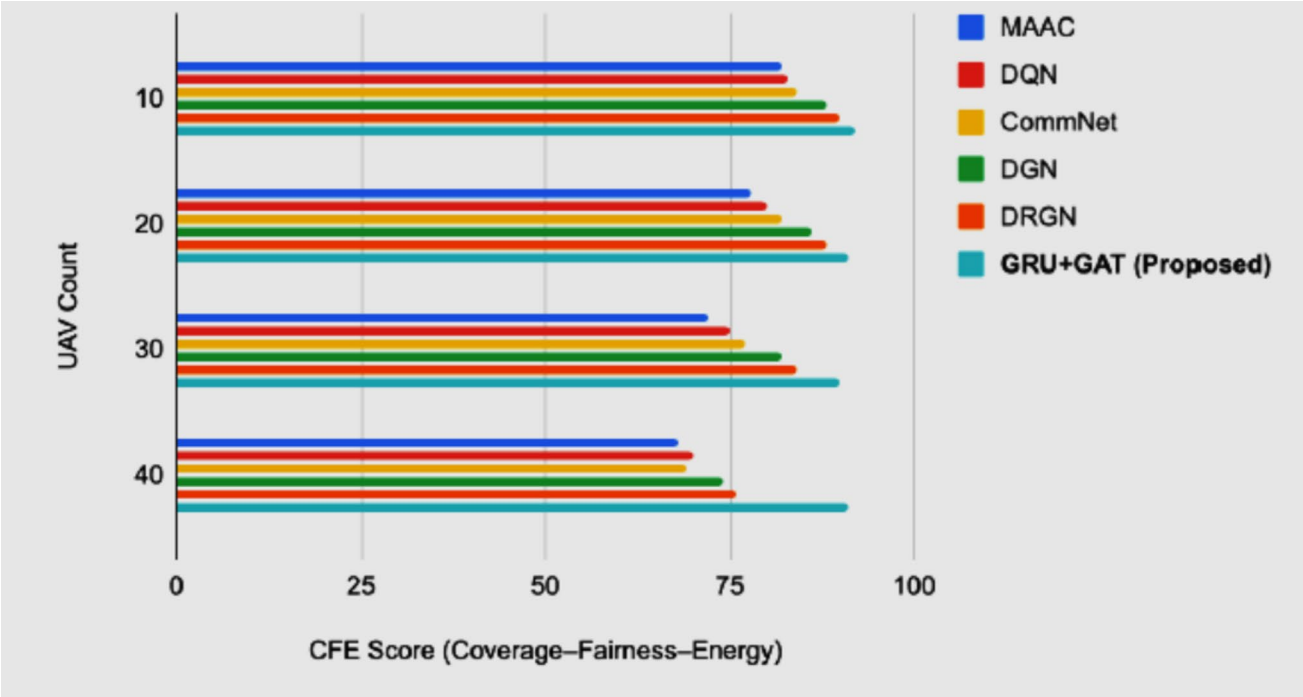

Fig. 4 CFE Score comparison

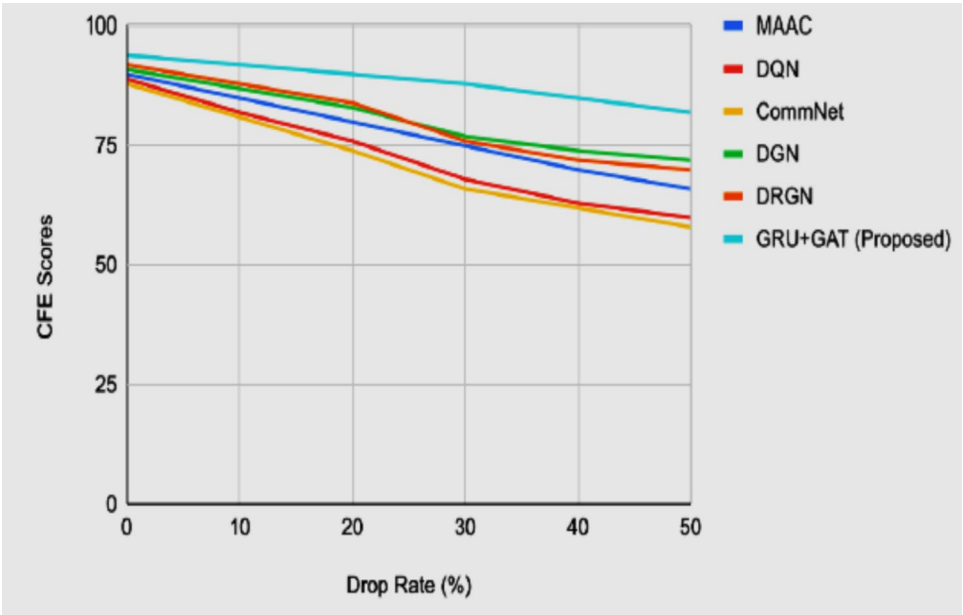

Fig. 5 Communication Drop Rate analysis

referenced UAV coordination strategies: Flying Ad-hoc Networks (FANETs) [7], ε-greedy exploration [8], game-theoretic models [9], mean-field control theory [10], multi-agent DRL (MADRL) [11], and centralized training with decentralized execution (CTDE) [12]. In raw accuracy, FANETs recorded 95.3%, ε-greedy recorded 96.1%, game-theoretic models recorded 97.0%, mean-field control theory recorded 97.4%, MADRL recorded 98.0%, and CTDE recorded 98.4%, but the proposed GRU + GAT surpassed all baselines with 99.2%.

While these absolute accuracy values are numerically close, a more distinct picture can be obtained when

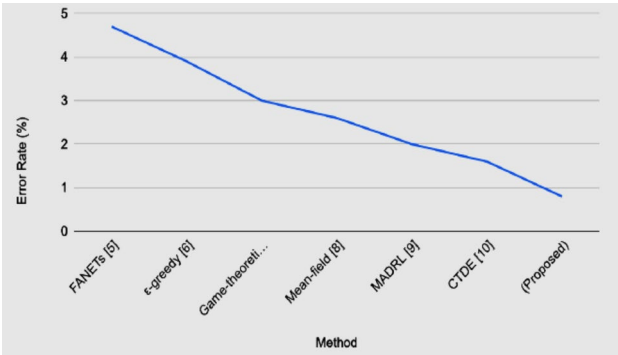

Fig. 6 Error rate comparison

converted to error rates (Fig. 6): FANETs are 4.7% errors, ε-greedy is 3.9%, game-theoretic models are 3.0%, mean-field control theory are 2.6%, MADRL are 2.0%, CTDE are 1.6%, and GRU + GAT is merely 0.8%. This way, the differences are consequential: GRU + GAT is twice as incorrect as CTDE and almost six times as incorrect as FANETs. Relative improvement once again emphasizes this trend, with ε-greedy decreasing errors by 17% relative to FANETs, game-theoretic by 36%, mean-field by 45%, MADRL by 57%, CTDE by 66%, and the proposed GRU + GAT by a remarkable 83%. The comparative point of view frames the benefit of the proposed method graphically clear, since its error bar is smaller than a half of the best next alternative and in proportion to legacy baselines. The superiority results from the synergy of spatial attention using GAT and temporal modeling using GRU, which collectively facilitate accurate decision-making under partially observable and dynamic UAV settings (Table 1).

5 Conclusion and future scope

This paper presents a fully decentralized DRL-based approach for UAV navigation in varying network conditions. It integrates GAT for self-attention mechanism and GRU-based memory units for temporal data retention, defined model effectively enables UAVs to collaborate and optimize communication coverage autonomously. Through extensive simulations, this study demonstrated that the proposed framework outperforms traditional methods in terms of navigation accuracy, and adaptability. The results show that defined model can successfully manage large UAV swarms and navigate complex, randomly distributed environments with partial observability constraints. The results show that the proposed GRU + GAT surpassed all baselines models with 99.2% navigation accuracy. Additionally, it has the ability to fine-tune the model in real-time which makes it highly practical for real-world deployment.

Future research can explore extending the proposed model to heterogeneous UAV swarms with varying capabilities, incorporating adaptive communication and investigating real-world implementation challenges. By further refining decentralized learning methodologies, UAV swarms can become even more effective in addressing critical real-world applications, such as emergency response and remote connectivity solutions.

Table 1 Validation of proposed work

Existing/proposed	Methods used	Navigation accuracy (%)	Error rate (%)
[7]	Flying Ad-hoc Networks (FANETs)	95.3	4.7
[8]	ε-greedy strategy	96.1	3.9
[9]	Game-theoretic frameworks	97.0	3.0
[10]	Mean-field game control theory	97.4	2.6
[11]	Multi-agent deep reinforcement learning (MADRL)	98.0	2.0
[12]	Centralized training with decentralized execution (CTDE)	98.4	1.6
Proposed	GAT + GRU	99.2	0.8

Declarations

Conflict of interest No competing interests associated with the given study.

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